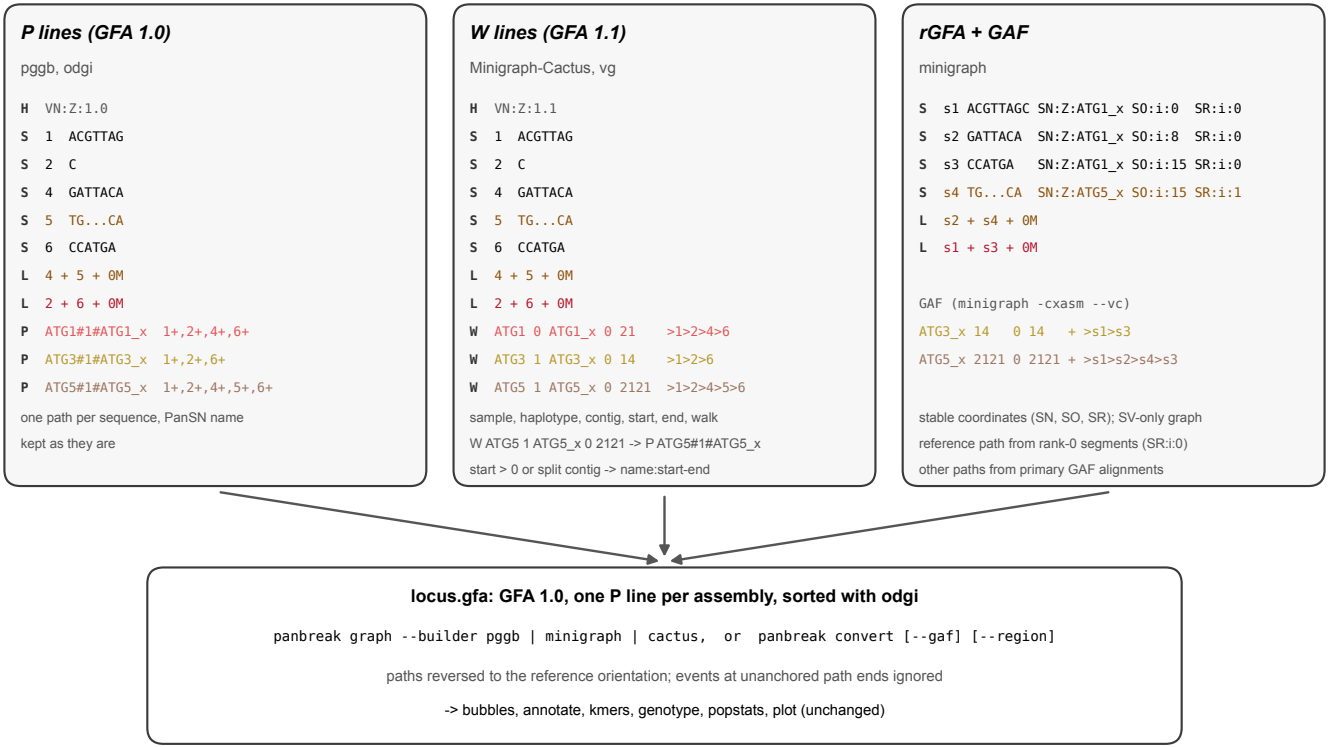

P lines (GFA 1.0)
pggb, odgi
H VN:Z:1.0
S 1 ACGTTAG
S 2 C
S 4 GATTACA
S 5 TG...CA
S 6 CCATGA
L 4 + 5 + 0M
L 2 + 6 + 0M
P ATG1#1#ATG1_x 1+,2+,4+,6+
P ATG3#1#ATG3_x 1+,2+,6+
P ATG5#1#ATG5_x 1+,2+,4+,5+,6+
one path per sequence, PanSN name
kept as they are
W lines (GFA 1.1)
Minigraph-Cactus, vg
H VN:Z:1.1
S 1 ACGTTAG
S 2 C
S 4 GATTACA
S 5 TG...CA
S 6 CCATGA
L 4 + 5 + 0M
L 2 + 6 + 0M
W ATG1 0 ATG1_x 0 21 >1>2>4>6
W ATG3 1 ATG3_x 0 14 >1>2>6
W ATG5 1 ATG5_x 0 2121 >1>2>4>5>6
sample, haplotype, contig, start, end, walk
W ATG5 1 ATG5_x 0 2121 -> P ATG5#1#ATG5_x
start > 0 or split contig -> name:start-end
rGFA + GAF
minigraph
S s1 ACGTTAGC SN:Z:ATG1_x SO:i:0 SR:i:0
S s2 GATTACA SN:Z:ATG1_x SO:i:8 SR:i:0
S s3 CCATGA SN:Z:ATG1_x SO:i:15 SR:i:0
S s4 TG...CA SN:Z:ATG5_x SO:i:15 SR:i:1
L s2 + s4 + 0M
L s1 + s3 + 0M
GAF (minigraph -cxasm --vc)
ATG3_x 14 0 14 + >s1>s3
ATG5_x 2121 0 2121 + >s1>s2>s4>s3
stable coordinates (SN, SO, SR); SV-only graph
reference path from rank-0 segments (SR:i:0)
other paths from primary GAF alignments
locus.gfa: GFA 1.0, one P line per assembly, sorted with odgi
panbreak graph --builder pggb | minigraph | cactus, or panbreak convert [--gaf] [--region]
paths reversed to the reference orientation; events at unanchored path ends ignored
-> bubbles, annotate, kmers, genotype, popstats, plot (unchanged)